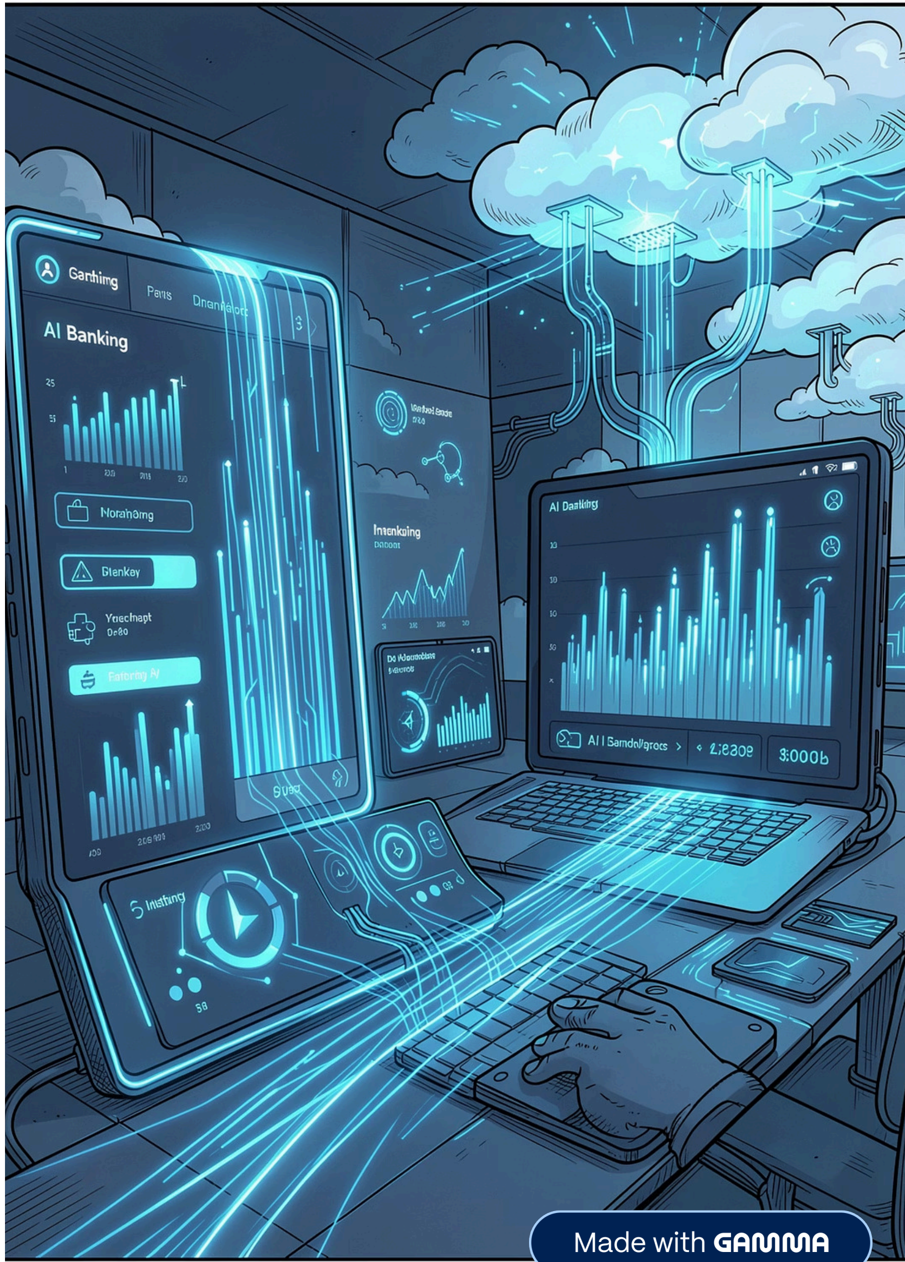
AI-Powered Bank Loan Prediction System - LoanIQ

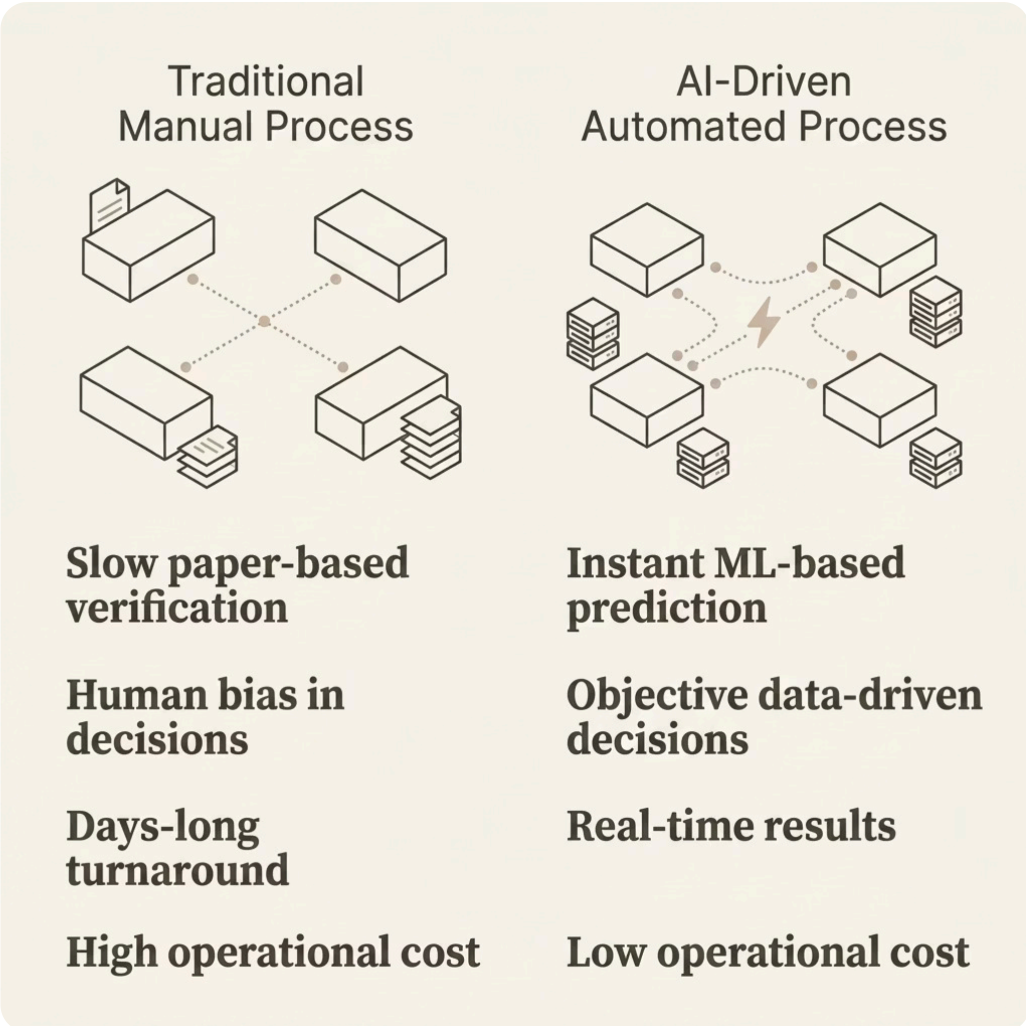
Machine Learning + Streamlit + AWS Cloud Deployment



Made with GAMMA

THE PROBLEM

Why This Project Matters



The Banking Gap

Traditional loan approval is slow, biased, and resource-intensive. Banks lose customers to faster competitors while manual verification introduces costly errors.

⏱️ Slow Processing

Days of manual review per application

⚖️ Human Bias

Inconsistent decisions across officers

💸 High Cost

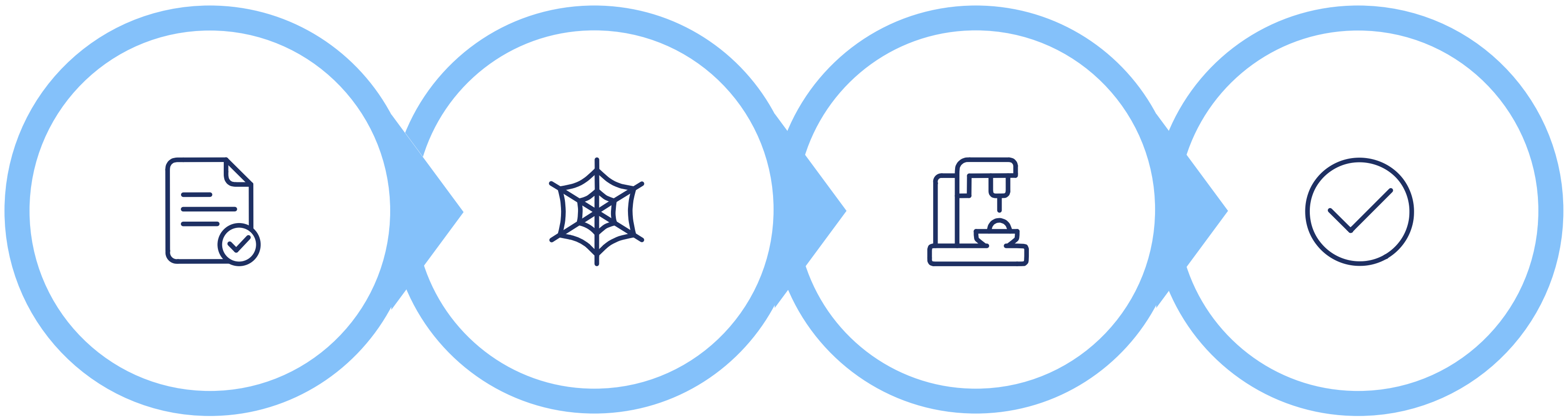
Operational overhead reduces margins

Made with GAMMA

THE SOLUTION

Proposed Solution

A cloud-deployed ML system that delivers instant, objective loan approval predictions through an intuitive web interface — eliminating manual bottlenecks and human bias.



Customer Inputs

Streamlit Interface

ML Prediction Engine

Loan Approval Result

The entire pipeline operates in real-time on AWS EC2, accessible globally via a public IP — no local installation required.

Made with GAMMA

FEATURES

Key Features

Real-Time Prediction

Instant loan decisions powered by a trained ML model — no waiting.

Interactive Streamlit UI

Clean, responsive web interface built entirely in Python.

AWS Cloud Deployment

Hosted on EC2 with SSH access, security groups, and public IP.

Scalable Architecture

Designed to scale from demo to production with minimal changes.

Consistent Decisions

ML eliminates human bias, ensuring fair and repeatable outcomes.

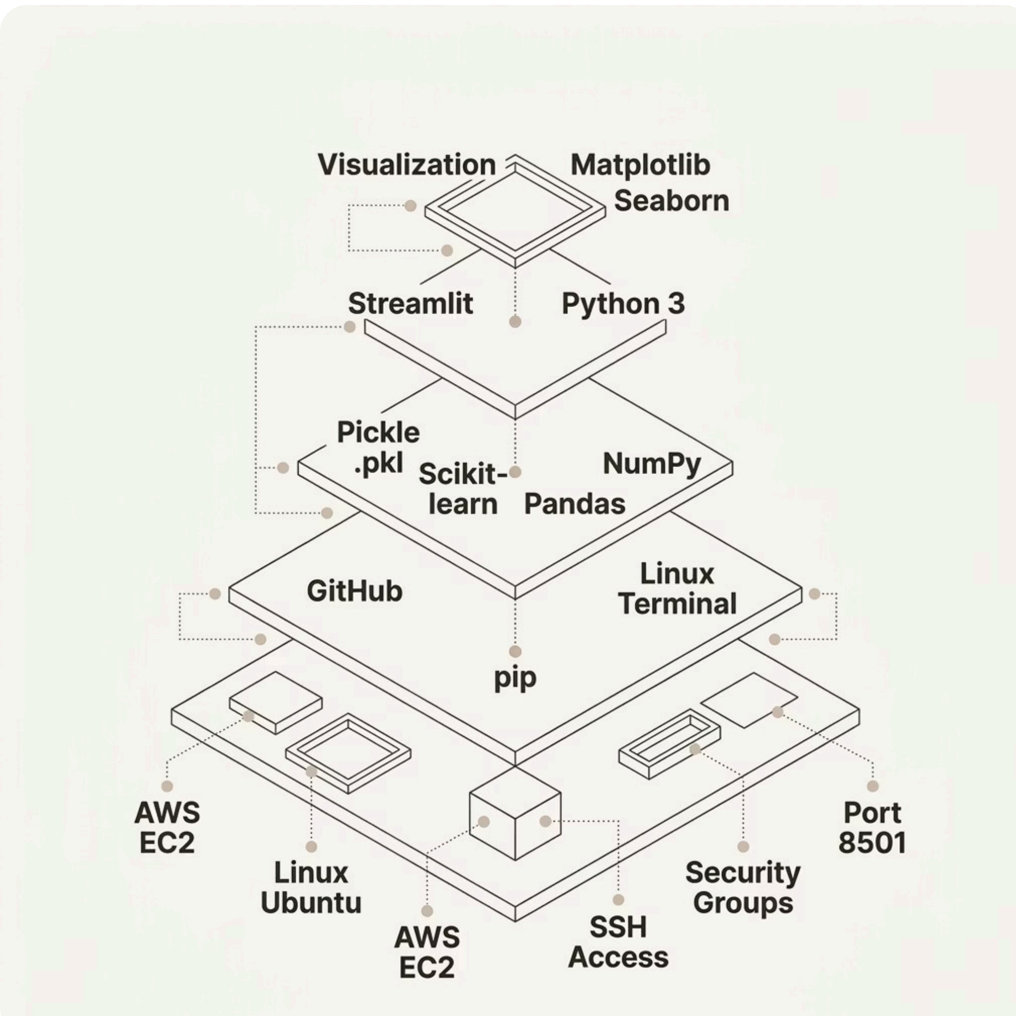
Open Source Ready

Full code on GitHub with requirements.txt and deployment guide.

Made with GAMMA

TECH STACK

Technology Stack



Stack Highlights

Every layer is open-source and cloud-native, enabling rapid iteration and zero licensing cost.

Python

Core language for ML and backend logic

Streamlit

Zero-friction frontend for data apps

Scikit-learn

Model training, preprocessing, evaluation

AWS EC2

Scalable virtual server with SSH access

Made with GAMMA

ML PIPELINE

Machine Learning Workflow

Data Collection

Loan dataset with key features

Data Cleaning

Handle missing values and encode

Feature Engineering

Select and create important attributes

Model Training

Train Scikit-learn classifier

Testing

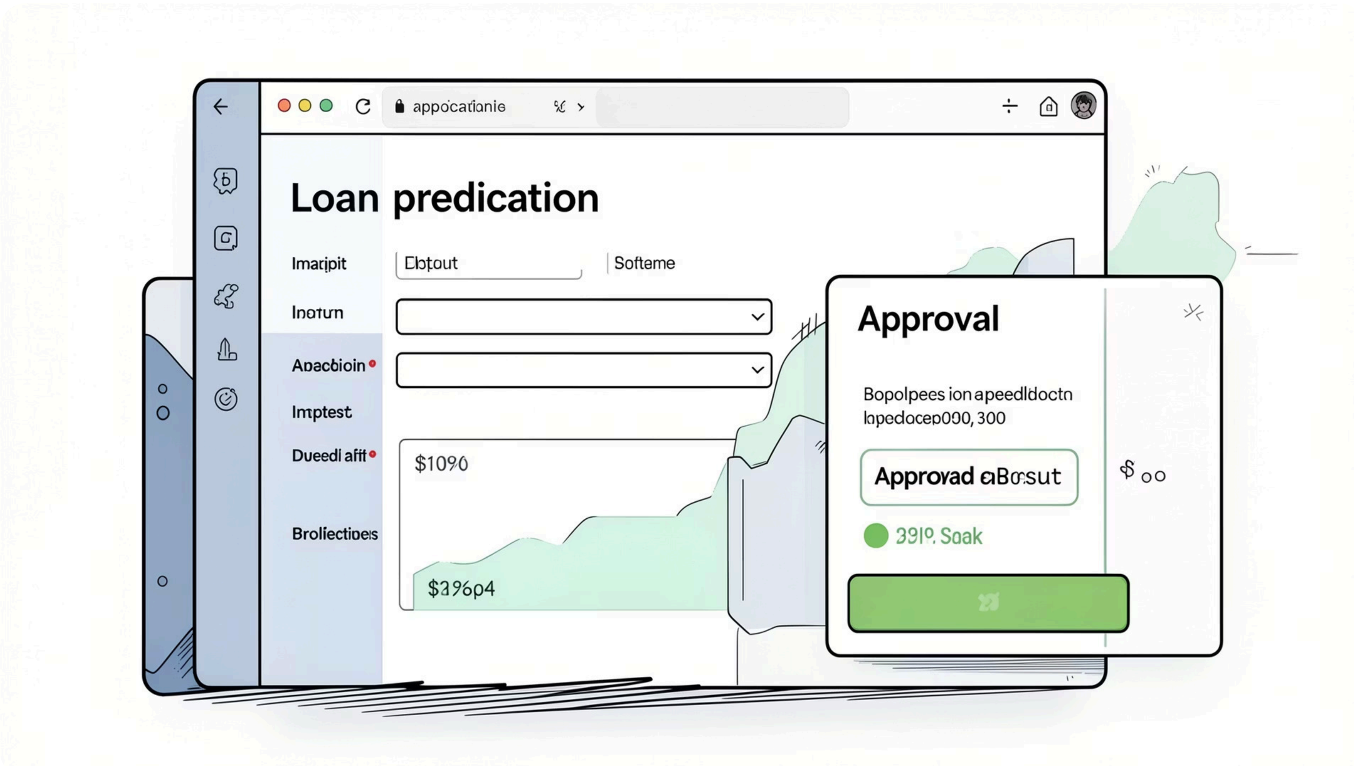
Evaluate accuracy and metrics

Key dataset features include **Applicant Income**, **Loan Amount**, **Credit History**, **Dependents**, **Education**, and **Property Area** — all contributing to a robust approval prediction.

Made with GAMMA

APPLICATION

Streamlit Application Design



How It Works

Streamlit renders a Python-driven frontend that captures user inputs, passes them to the serialized ML model, and displays the prediction result — all within a single session.

1 **Input Form**

User enters income, loan amount, credit history, and other attributes

2 **Backend Processing**

Python logic preprocesses inputs and calls `ML_Model.pkl`

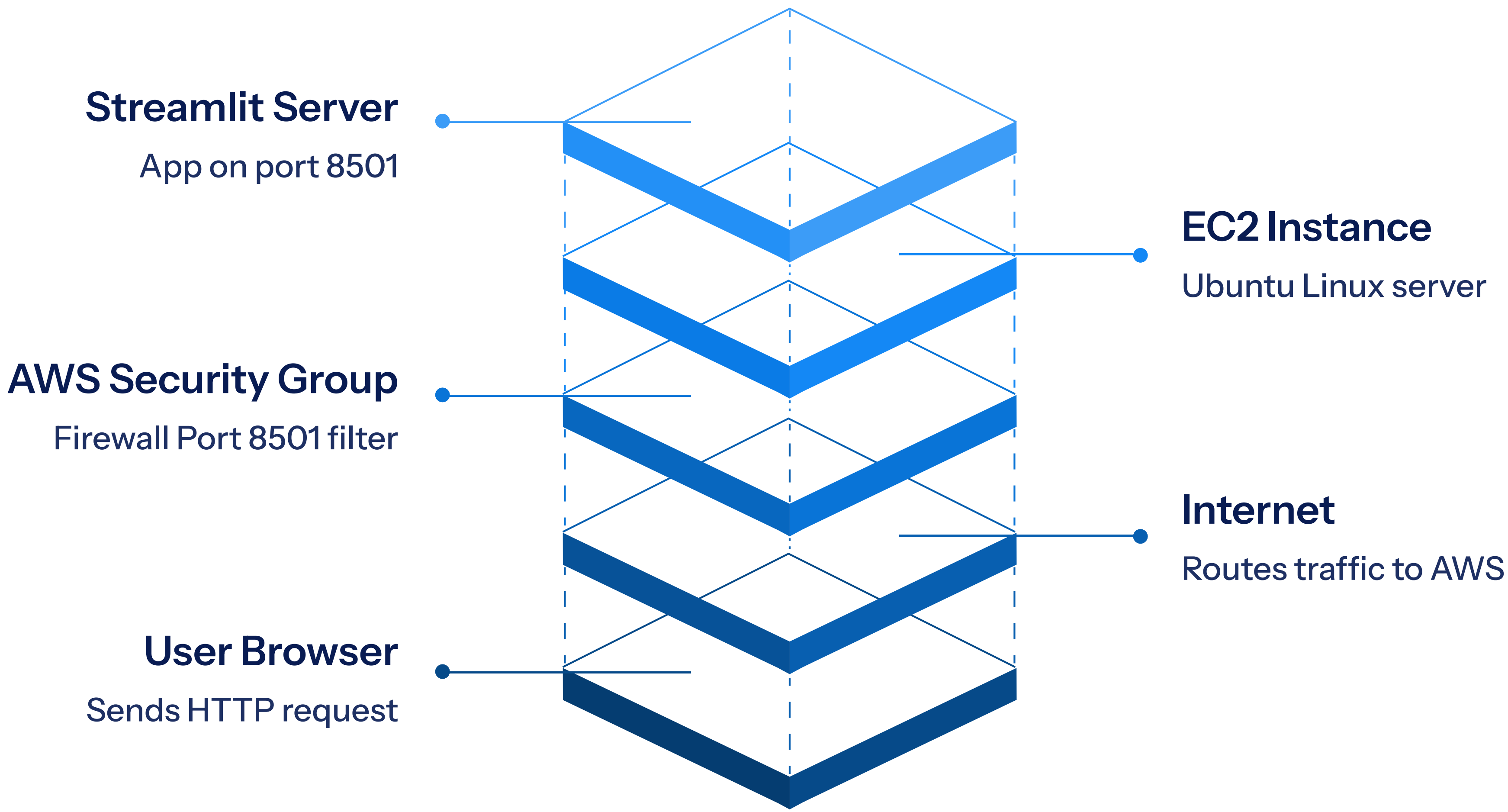
3 **Result Display**

Approved / Rejected shown instantly with confidence indicators

Made with GAMMA

ARCHITECTURE

Complete AWS System Architecture



Every request traverses AWS Security Groups before reaching the EC2 instance. The Streamlit server runs on **Port 8501**, serving predictions from the serialized model. Developers maintain the system via **SSH** and push updates through **GitHub**.

Made with GAMMA

DEPLOYMENT

AWS Deployment Workflow

01

Launch EC2 Instance

Ubuntu 20.04, t2.micro or higher

02

Configure Security Groups

Open Port 8501 for Streamlit

03

SSH into Server

`ssh -i key.pem ubuntu@public-ip`

04

Install Python & pip

`sudo apt install python3-pip`

05

Clone GitHub Repo

`git clone https://github.com/Spidy20/Streamlit_Bank_Loan_Prediction`

06

Install Dependencies

`pip install -r requirements.txt`

07

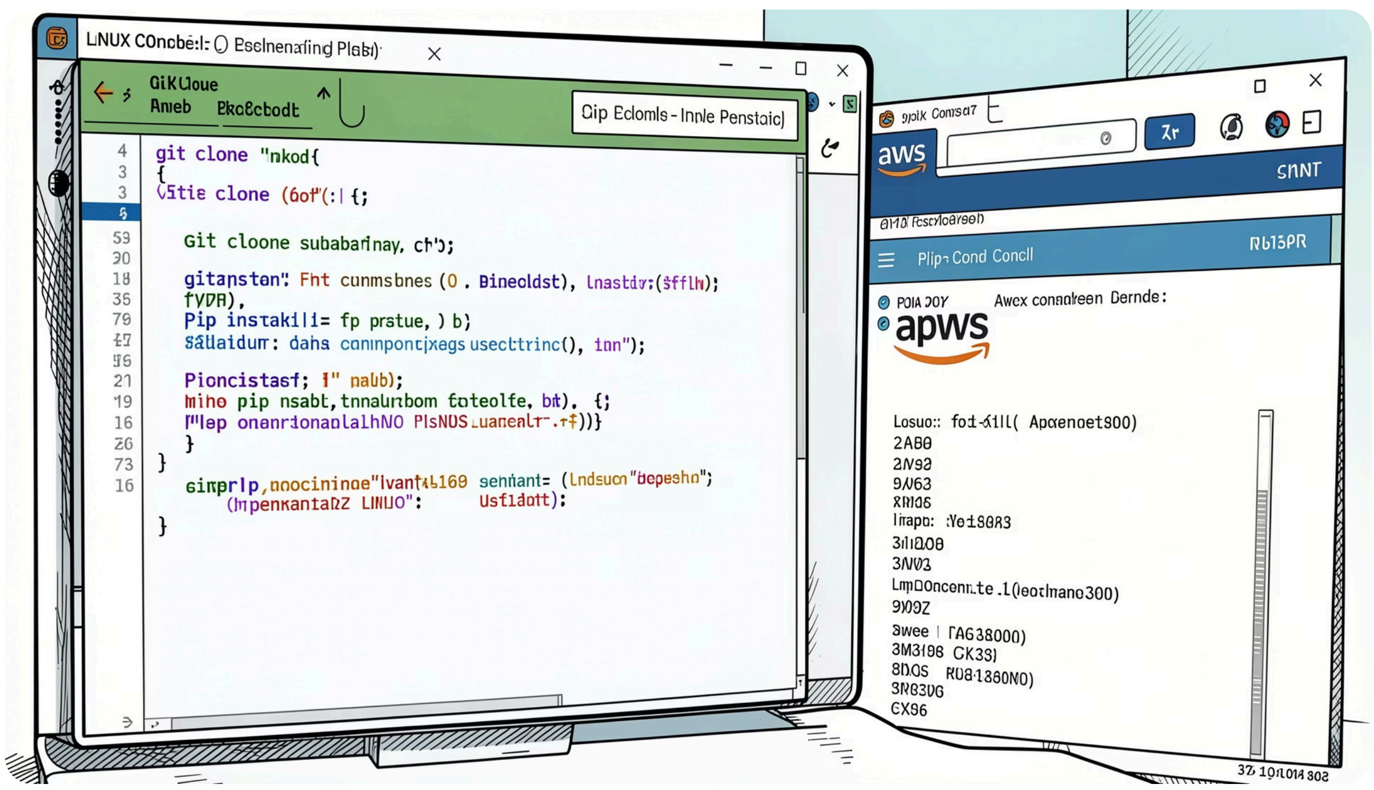
Start Streamlit Server

`streamlit run Bank_Loan_Prediction.py`

08

Access via Public IP

Open `http://public-ip:8501` in browser



Made with GAMMA

CHALLENGES

Challenges & Debugging

⚠️ Missing Python Modules

Resolved via `pip install -r requirements.txt`

⚠️ .pkl Model Not Found

Ensured file path matches working directory on EC2

⚠️ Security Group Blocking

Opened Port 8501 in AWS Console inbound rules

⚠️ scikit-learn Version Mismatch

Pinned exact versions in `requirements.txt`

Made with GAMMA